№ 13

$\beta = 0{,}95 \Rightarrow \alpha = 0{,}05$

	Египет	Европа
n/m	139	1000
Длина	5,722	6,161
Ширина	4,612	5,055

а) длина

$H_0: \sigma_{ег}^2 = \sigma_{ев}^2$

$H_1: \sigma_{ег}^2 \neq \sigma_{ев}^2$

$S_{ев}^2 = \frac{m}{m-1}\sigma_{ев}^2 \approx 37{,}99592$

$S_{ег}^2 = \frac{n}{n-1}\sigma_{ег}^2 \approx 32{,}97854$

$\tilde{\Delta} = \frac{S_{ег}^2}{S_{ев}^2} \approx 0{,}867950$

$\text{p-value} = P(\Delta \geq \tilde{\Delta} \mid H_0) = \int_{0{,}86795}^{+\infty} q(t)\,dt \approx 0{,}875$

$\frac{\alpha}{2} = 0{,}025 \qquad 1 - \frac{\alpha}{2} = 0{,}975$

$0{,}025 < 0{,}875 < 0{,}975 \Rightarrow$

$\Rightarrow$ нет оснований отвергать H_0

б) ширина

$H_0: \sigma_{er}^2 = \sigma_{eb}^2$

$H_1: \sigma_{er}^2 \neq \sigma_{eb}^2$

$S_{eb}^2 = \frac{m}{m-1}\sigma_{eb}^2 \approx 25,57860$

$S_{er}^2 = \frac{n}{n-1}\sigma_{er}^2 \approx 21,42468$

$\hat{\Delta} = \frac{S_{er}^2}{S_{eb}^2} \approx 0,837602$

$p\text{-}value = P(\Delta \geq \hat{\Delta} \mid H_0) = \int_{0,837602}^{+\infty} q(t)dt \approx 0,9197$

$0,025 < 0,9197 < 0,975 \Rightarrow$

$\Rightarrow$ нет оснований отвергать H_0

в) $W(\theta) = P(\Delta \geq \hat{\Delta} \mid H_1) =$

$= P\left(\frac{S_{er}^2}{S_{eb}^2} \geq u_{1-\frac{\alpha}{2}} \mid H_1\right) + P\left(\frac{S_{er}^2}{S_{eb}^2} \leq u_{\frac{\alpha}{2}} \mid H_1\right) =$

$= P\left(\frac{S_{er}^2 \cdot \sigma_{eb}^2}{\sigma_{er}^2 \cdot S_{eb}^2} > \underbrace{\frac{\sigma_{eb}^2}{\sigma_{er}^2} \cdot u_{1-\frac{\alpha}{2}}}_{a_1(\theta)}\right) + P\left(\frac{S_{er}^2 \cdot \sigma_{eb}^2}{S_{eb}^2 \cdot \sigma_{er}^2} < \underbrace{u_{\frac{\alpha}{2}} \cdot \frac{\sigma_{eb}^2}{\sigma_{er}^2}}_{a_2(\theta)}\right) =$

$= \int_{a_1}^{+\infty} q(t)dt + \int_0^{a_2} q(t)dt = 1 - F(a_2) + F(a_1) \Rightarrow$

$\Rightarrow W(\theta) = 1 - F(a_2(\theta)) + F(a_1(\theta))$

$u_{\frac{\alpha}{2}} \approx 0,767 \qquad u_{1-\frac{\alpha}{2}} \approx 1,272$

График мощности критерия by Юрий